

Sentinel Fraud Detection — API Reference

Base URL: <https://api.freud.dpdns.org>

All requests and responses use JSON. No authentication required.

Interactive Docs (Swagger UI)

```
https://api.freud.dpdns.org/accounts/docs
https://api.freud.dpdns.org/transactions/docs
```

Endpoints

GET /health

Check if the API is up.

Response

```
{ "status": "ok" }
```

POST /accounts

Create a new account.

Request Body

Field	Type	Required	Description
phone_number	string	Yes	E.164 format — must include country code
home_state	string	Yes	2-letter US state code
dob	string	Yes	Date of birth in YYYY-MM-DD format
threshold	float	No	Fraud sensitivity 0.0–1.0, defaults to 0.8

Example Request

```
{
  "phone_number": "+16085551234",
  "home_state": "WI",
  "dob": "1998-01-07",
  "threshold": 0.8
}
```

Example Response — 200

```
{ "account_id": "b3f1c2d4-8e6a-4f2b-a1c3-d4e5f6a7b8c9" }
```

Error Responses

Status	Reason
400	Invalid phone_number format (must be E.164, e.g. +16085551234)
400	Invalid home_state (must be a valid 2-letter US state code)
400	Invalid dob format (must be YYYY-MM-DD)
409	An account with this phone number already exists

GET /accounts/by-phone/{phone_number}

Retrieve an account and its transaction history.

URL Parameter

Param	Description
phone_number	E.164 phone number — URL-encode the + as %2B

Query Parameter

Param	Type	Default	Description
limit	int	20	Max number of transactions to return

Example Request

```
GET /accounts/by-phone/%2B16085551234?limit=10
```

Example Response — 200

```
{
  "account": {
    "account_id": "b3f1c2d4-...",
    "phone_number": "+16085551234",
    "home_state": "WI",
    "dob": "1998-01-07",
    "threshold": 0.8,
    "created_at": "2026-04-19T12:00:00+00:00"
  },
  "transactions": [
    {
      "transaction_id": "a1b2c3d4-...",
      "amount": 500.00,
      "merchant": "Amazon",
      "category": "shopping",
      "state": "CA",
      "score": 0.87,
      "is_fraud": false,
      "pending_confirm": 0,
      "created_at": "2026-04-19T13:00:00+00:00"
    }
  ]
}
```

Error Responses

Status	Reason
404	Account not found

PATCH /accounts/by-phone/{phone_number}/threshold

Update the fraud sensitivity threshold for an account.

Request Body

Field	Type	Required	Description
threshold	float	Yes	New sensitivity value, 0.0–1.0

Example Request

```
{ "threshold": 0.6 }
```

Example Response — 200

```
{
  "account_id": "b3f1c2d4-...",
  "phone_number": "+16085551234",
  "old_threshold": 0.8,
  "new_threshold": 0.6,
  "updated_at": "2026-04-19T14:00:00+00:00"
}
```

Error Responses

Status	Reason
404	Account not found

DELETE /accounts/by-phone/{phone_number}

Delete an account.

Example Response — 200

```
{ "deleted": "b3f1c2d4-8e6a-4f2b-a1c3-d4e5f6a7b8c9" }
```

Error Responses

Status	Reason
404	Account not found

POST /transactions

Submit a transaction to be scored for fraud.

Request Body

Field	Type	Required	Description
phone_number	string	Yes	E.164 format — used to look up the account
amount	float	Yes	Transaction amount, must be greater than 0
merchant	string	Yes	Merchant name
category	string	Yes	Transaction category
state	string	Yes	2-letter US state where transaction occurred

Example Request

```
{
  "phone_number": "+16085551234",
  "amount": 500.00,
  "merchant": "Amazon",
  "category": "shopping",
  "state": "CA"
}
```

Example Response — 200

```
{
  "transaction_id": "a1b2c3d4-5e6f-7a8b-9c0d-e1f2a3b4c5d6",
  "score": 0.87,
  "flagged": true
}
```

Response Fields

Field	Description
transaction_id	UUID for this transaction
score	Fraud percentile 0.0–1.0. Score of 0.87 = more suspicious than 87% of all transactions
flagged	true if score exceeded the account threshold — WhatsApp alert has been sent

Note: The first request after a cold start may take 5–10 seconds while the ML model loads from S3.

Error Responses

Status	Reason
404	No account found for this phone number

Understanding the Threshold

Controls fraud sensitivity per account. Set at creation, updatable anytime via PATCH.

Threshold	Behavior
0.9	Only alert on top 10% most suspicious — fewer alerts, may miss some fraud
0.8	Default — good balance between catching fraud and false positives
0.5	Alert on anything above median suspicion
0.3	Very sensitive — catches more fraud but more false positives

Recommended UI: expose as a slider (0.0–1.0) with a default of 0.8.

WhatsApp Confirmation Flow

When a transaction is flagged, the following happens automatically:

Step	What Happens
1	Cardholder receives WhatsApp message with amount, merchant, state, and time
2 — YES	Transaction marked as not fraud — <code>is_fraud = false</code> , <code>pending_confirm = 0</code>
3 — NO	Transaction marked as fraud — <code>is_fraud = true</code> , <code>pending_confirm = 0</code>

pending_confirm is 1 while awaiting a reply, 0 after the cardholder responds.

Phone Number Formatting

All phone numbers must be in E.164 format — include country code, start with +, no spaces or dashes.

US example: +16085551234

In URL paths, encode + as %2B:
`/accounts/by-phone/%2B16085551234`